ROTTAY PLATFORM - Technical Reference | January 2026

> ***626,862 lines of code | 731 use cases | 156 entities | 734 test files***
>
> ***Full-Stack Platform: Backend + Frontend + Infrastructure***

A multi-tenant SaaS platform built with Hexagonal Architecture, Domain-Driven Design, and CQRS patterns. Production-ready infrastructure supporting multiple verticals across talent management, hospitality, events, Web3, and AI-powered services.

Rottay is not just backend modules — it's a complete full-stack platform including:

- **Backend**: 15 modules with 731 use cases
- **Frontend**: Design system with 4 UI engines and customizable themes
- **Infrastructure**: Multi-tenant architecture, auto-scaling, CI/CD, and monitoring

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1. Overview

Technical Philosophy

The Rottay platform is engineered around three core principles:

1. **Domain Isolation** - Each business domain operates as an independent module with zero coupling to siblings
2. **Infrastructure Agnosticism** - Business logic has no knowledge of databases, APIs, or external services
3. **Explicit Over Implicit** - Every dependency is injected, every operation returns a typed Result

Architecture: Hexagonal + DDD + CQRS

```
+-----------------------------------------------------------------+
|                     HEXAGONAL ARCHITECTURE                       |
+-----------------------------------------------------------------+
|                                                                  |
|  +------------------+     +------------------+                   |
|  |   PRIMARY PORTS  |     |  SECONDARY PORTS |                   |
|  |   (Driving Side) |     |  (Driven Side)   |                   |
|  +------------------+     +------------------+                   |
|           |                         ^                            |
|           v                         |                            |
|  +------------------------------------------+                    |
|  |             APPLICATION CORE              |                   |
|  |  +--------------------------------------+ |                   |
|  |  |           USE CASES (CQRS)           | |                   |
|  |  |  +----------------+  +-------------+ | |                   |
|  |  |  |    COMMANDS    |  |   QUERIES   | | |                   |
|  |  |  | (State Changes) |  | (Read-Only) | | |                  |
|  |  |  +----------------+  +-------------+ | |                   |
|  |  +--------------------------------------+ |                   |
|  |  +--------------------------------------+ |                   |
|  |  |             DOMAIN LAYER             | |                   |
|  |  |  Entities | Value Objects | Aggregates | |                 |
|  |  +--------------------------------------+ |                   |
|  +------------------------------------------+                    |
|           ^                         |                            |
|           |                         v                            |
|  +------------------+     +------------------+                   |
|  |     ADAPTERS     |     |     ADAPTERS     |                   |
|  |   (Controllers)  |     |  (Repositories)  |                   |
|  +------------------+     +------------------+                   |
|                                                                  |
+-----------------------------------------------------------------+
```

Why This Matters

Benefit	Implementation
Testability	Domain logic tested without infrastructure
Flexibility	Swap PostgreSQL for MongoDB without touching business code
Maintainability	15 modules, each with clear boundaries and ownership
Scalability	Extract any module to microservice without refactoring
Onboarding	Consistent patterns across 626K+ lines of code

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2. Architecture at a Glance

```
+=======================================================================+
|                         ROTTAY PLATFORM ARCHITECTURE                    |
+=======================================================================+

                          +-------------------------+
                          |        VERTICALS        |
                          |  BITHIRE | NOCTIS | etc. |
                          +-------------------------+
                                       |
                                       v
                    +-----------------------------------+
                    |     PORTAL (Next.js 15 + RSC)     |
                    |  App Router | Server Actions      |
                    +-----------------------------------+
                                       |
                    +------------------+------------------+
                    |                  |                  |
                    v                  v                  v
    +----------------+    +----------------+    +----------------+
    | @rottay/auth   |    | @rottay/tenancy|    | @rottay/perms  |
    | @rottay/identity    | @rottay/flags  |    | @rottay/nav    |
    |      +------------------------------------------+         |
    |      |           PLATFORM MODULES (7)           |         |
    |      |           78 + 50 + 22 + 28 + 22 + 30 + 40         |
    |      |              = 270+ Use Cases            |         |
    +------+------------------------------------------+---------+
                                       |
                    +------------------+------------------+
                    |                  |                  |
                    v                  v                  v
    +----------------+    +----------------+    +----------------+
    | @rottay/talent |    | @rottay/staff  |    | @rottay/web3   |
    | @rottay/assess |    | @rottay/bar    |    | @rottay/pay    |
    |      +------------------------------------------+         |
    |      |            DOMAIN MODULES (8)            |         |
    |      |     89+76+183+60+98+82+114+29 = 731      |         |
    |      |              Use Cases                   |         |
    +------+------------------------------------------+---------+
                                       |
                                       v
                    +-----------------------------------+
                    |           @rottay/core            |
                    |           450+ Exports            |
                    | Errors | Base Classes | Utils     |
                    +-----------------------------------+
                                       |
             +------------+------------+------------+------------+
             |            |            |            |            |
             v            v            v            v            v
    +-----------+  +-----------+  +---------+  +----------+  +------------+
    | PostgreSQL|  |   Redis   |  | Typesense |  |   SQS   |  | S3 + CDN  |
    |   (RDS)   |  | (Cluster) |  | Search  |  | Queue  |  |   Assets   |
    +-----------+  +-----------+  +---------+  +----------+  +------------+
```

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3. Tech Stack

Layer	Technology	Version	Purpose
Runtime	Node.js	20 LTS	Server runtime with native ESM
Language	TypeScript	5.4+	Strict mode, decorators enabled
Framework	Next.js	15	App Router, RSC, Server Actions
ORM	Drizzle	0.30+	Type-safe queries, migrations
Database	PostgreSQL	15	Primary datastore (AWS RDS)
Cache	Redis	7	Session store, rate limiting, pub/sub
Search	Typesense	0.25	Full-text search, faceted filtering
Queue	AWS SQS	-	Async job processing
Storage	AWS S3	-	File uploads, assets
CDN	CloudFront	-	Static asset delivery
Infra	AWS	-	ECS, RDS, ElastiCache, SQS, S3
Monorepo	Turborepo	2.0	Build orchestration, caching
Testing	Vitest	1.0+	Unit, integration, coverage
E2E	Playwright	1.40+	Browser automation
Validation	Zod	3.22+	Runtime type validation

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4. @rottay/core

> ***Server-only foundation package with 450+ exports***

```
// Package is marked server-only - cannot be imported in client components
import "server-only";
```

Exports by Category

Category	Count	Key Exports
Error Classes	25+	`DomainError`, `ValidationError`, `NotFoundError`, `ConflictError`, `UnauthorizedError`, `ForbiddenError`, `InfrastructureError`, `ExternalServiceError`
Base Classes	15+	`Entity`, `AggregateRoot`, `ValueObject`, `UseCase`, `Repository`, `Mapper`
Database	30+	`DrizzleClient`, `Transaction`, `withTransaction`, `TenantContext`, `AuditFields`
Validation	40+	`validate`, `validateAsync`, `createValidator`, `schemas.*`
Utilities	50+	`generateId`, `generateSlug`, `hashPassword`, `verifyPassword`, `encrypt`, `decrypt`
Result Pattern	10+	`Result`, `success`, `error`, `isSuccess`, `isError`, `unwrap`, `unwrapOr`
Types	100+	`TenantId`, `UserId`, `EntityId`, `Timestamp`, `Money`, `Email`, `Phone`
Decorators	20+	`@Encrypted`, `@Indexed`, `@Unique`, `@Audited`, `@Cached`
Constants	50+	`ERROR_CODES`, `HTTP_STATUS`, `CACHE_TTL`, `PAGINATION`
Helpers	60+	`paginate`, `sort`, `filter`, `buildQuery`, `parseFilters`

Result Pattern Implementation

```
// Every use case returns Result<T, E>
type Result<T, E = DomainError> = Success<T> | Failure<E>;

// Usage in use cases
export class CreateUserUseCase {
  async execute(input: CreateUserInput): Promise<Result<User>> {
    const validation = validate(createUserSchema, input);
    if (isError(validation)) {
      return error(new ValidationError(validation.error));
    }

    const existingUser = await this.userRepo.findByEmail(input.email);
    if (existingUser) {
      return error(new ConflictError("User already exists"));
    }

    const user = await this.userRepo.create(input);
    return success(user);
  }
}
```

Quick Start Example

```
// Create a user with @rottay/auth
import { makeCreateUserUseCase, isSuccess } from "@rottay/auth";
import { makeTenantContext } from "@rottay/tenancy";

// Initialize use case with dependencies
const createUser = makeCreateUserUseCase({ db, logger });

// Create tenant context from JWT
const ctx = makeTenantContext({ tenantId: "tenant_123", userId: "admin_456" });

// Execute use case
const result = await createUser.execute(ctx, {
  email: "user@example.com",
  password: "securePassword123",
  role: "member"
});

if (isSuccess(result)) {
  console.log("User created:", result.data.id);
} else {
  console.error("Error:", result.error.message);
}
```

---

5. Platform Modules

> ***7 modules providing cross-cutting concerns for all verticals***

Module Overview

Module	Use Cases	Purpose
@rottay/auth	78	Authentication, sessions, MFA
@rottay/tenancy	50+	Multi-tenant isolation
@rottay/identity	22+	User profiles, teams
@rottay/permissions	28+	RBAC, resource access
@rottay/feature-flags	22+	Feature management
@rottay/navigation	30+	Dynamic menus, guards
@rottay/compliance	40+	Regulatory compliance

@rottay/auth (78 Use Cases)

```
+-----------------------------------------------------------------+
|                      AUTHENTICATION MODULE                       |
+-----------------------------------------------------------------+

  REGISTRATION (12)         LOGIN (15)              MFA (18)
  ----------------          ---------               --------
  - RegisterUser            - LoginWithEmail        - EnableMFA
  - VerifyEmail             - LoginWithPhone        - DisableMFA
  - ResendVerification      - LoginWithOAuth        - GenerateTOTP
  - CompleteOnboarding      - ValidateCredentials   - VerifyTOTP
  - CreateTenantAdmin       - RefreshToken          - GenerateBackupCodes
  - InviteUser              - ImpersonateUser       - ValidateBackupCode
  - AcceptInvitation        - SwitchTenant          - SendSMSCode
  - SetInitialPassword      - ValidateSession       - VerifySMSCode
  - UpdatePassword          - ExtendSession         - RegisterSecurityKey
  - ValidatePasswordStrength- RevokeToken           - ValidateSecurityKey
  - CheckEmailAvailability  - LogoutUser            - ListMFAMethods
  - CheckUsernameAvailable  - LogoutAllDevices      - SetPreferredMFA
                            - GetActiveDevices      - RequireMFAForRole
                            - TrustDevice           - AuditMFAAttempt
                            - RevokeDeviceTrust     - ResetMFA

  SESSIONS (14)             OAUTH (10)              RECOVERY (9)
  ------------              --------                ----------
  - CreateSession           - InitiateOAuth         - RequestPasswordReset
  - ValidateSession         - HandleCallback        - ValidateResetToken
  - RefreshSession          - LinkProvider          - ResetPassword
  - TerminateSession        - UnlinkProvider        - RequestAccountRecovery
  - ListUserSessions        - GetLinkedProviders    - ValidateRecoveryCode
  - TerminateAllSessions    - RefreshOAuthToken     - RecoverAccount
  - GetSessionMetadata      - ValidateOAuthState    - GenerateRecoveryCodes
  - UpdateSessionActivity   - ExchangeCode          - ListRecoveryOptions
  - SetSessionTimeout       - GetUserInfo           - AuditRecoveryAttempt
  - EnforceSessionLimits    - RevokeOAuthAccess
  - DetectConcurrentSession
  - HandleSessionConflict
  - MigrateSession
  - AuditSessionEvent
```

@rottay/tenancy (50+ Use Cases)

```
  TENANT CRUD (15)          ISOLATION (12)          BRANDING (10)
  ---------------           -------------           ------------
  - CreateTenant            - EnforceTenantScope    - SetTenantBranding
  - UpdateTenant            - ValidateTenantAccess  - UpdateLogo
  - DeleteTenant            - SwitchTenantContext   - SetColorScheme
  - GetTenant               - GetTenantMetadata     - ConfigureEmailTemplate
  - ListTenants             - CloneTenantConfig     - SetCustomDomain
  - ArchiveTenant           - MergeTenants          - ConfigureSSOBranding
  - RestoreTenant           - SplitTenant           - SetFavicon
  - TransferOwnership       - ValidateDataAccess    - UpdateLoginPage
  - UpdateTenantSettings    - AuditCrossAccess      - SetWatermark
  - SetTenantLimits         - EnforceRowSecurity    - ExportBrandAssets
  - GetTenantUsage          - ScopedQueryBuilder
  - SuspendTenant           - ValidateTenantId
  - ReactivateTenant
  - SetTenantPlan
  - MigrateTenantData

  PROVISIONING (13+)
  -----------------
  - ProvisionTenant
  - ProvisionDatabase
  - SeedTenantData
  - ConfigureIntegrations
  - SetupBilling
  - CreateAdminUser
  - SendWelcomeEmail
  - ValidateProvisioning
  - RollbackProvisioning
  - ScheduleProvisioning
  - BulkProvision
  - CloneTenant
  - ImportTenantData
```

@rottay/identity (22+ Use Cases)

```
  PROFILES (8)              TEAMS (7)               ORGANIZATIONS (7+)
  -----------               --------                -----------------
  - CreateProfile           - CreateTeam            - CreateOrganization
  - UpdateProfile           - UpdateTeam            - UpdateOrganization
  - GetProfile              - DeleteTeam            - GetOrganization
  - DeleteProfile           - AddTeamMember         - ListOrganizations
  - UploadAvatar            - RemoveTeamMember      - InviteToOrganization
  - SetProfileVisibility    - GetTeamMembers        - AcceptOrgInvitation
  - LinkSocialProfile       - TransferTeamOwner     - LeaveOrganization
  - VerifyIdentity
```

@rottay/permissions (28+ Use Cases)

```
  ROLES (10)                PERMISSIONS (10)        HIERARCHIES (8+)
  ---------                 ---------------         ---------------
  - CreateRole              - GrantPermission       - CreateHierarchy
  - UpdateRole              - RevokePermission      - UpdateHierarchy
  - DeleteRole              - CheckPermission       - DeleteHierarchy
  - AssignRole              - ListPermissions       - AssignToHierarchy
  - RevokeRole              - BulkGrant             - GetAncestors
  - ListRoles               - BulkRevoke            - GetDescendants
  - GetRolePermissions      - GetEffectivePerms     - ValidateHierarchy
  - CloneRole               - InheritPermissions    - FlattenHierarchy
  - SetDefaultRole          - AuditPermission
  - ValidateRoleAssignment  - ScopePermission
```

@rottay/feature-flags (22+ Use Cases)

```
  BOOLEAN FLAGS (6)         ROLLOUT (6)             A/B TESTING (5)
  ----------------          ----------              --------------
  - CreateFlag              - SetRolloutPercentage  - CreateExperiment
  - UpdateFlag              - IncrementRollout      - AssignVariant
  - DeleteFlag              - PauseRollout          - RecordConversion
  - EvaluateFlag            - ResumeRollout         - GetExperimentResults
  - ListFlags               - GetRolloutStatus      - EndExperiment
  - BulkEvaluate            - ScheduleRollout

  TARGETING (5+)
  -------------
  - CreateTargetingRule
  - EvaluateTargeting
  - SetUserOverride
  - SetTenantOverride
  - ClearOverrides
```

@rottay/navigation (30+ Use Cases)

```
  MENUS (10)                BREADCRUMBS (6)         GUARDS (8)
  ---------                 --------------          ---------
  - CreateMenu              - GenerateBreadcrumbs   - RegisterGuard
  - UpdateMenu              - GetBreadcrumbPath     - EvaluateGuard
  - DeleteMenu              - SetBreadcrumbLabel    - ChainGuards
  - AddMenuItem             - CacheBreadcrumbs      - BypassGuard
  - RemoveMenuItem          - ResolveDynamicCrumb   - AuditGuardResult
  - ReorderMenuItems        - GetParentCrumbs       - SetGuardPriority
  - SetMenuVisibility                               - ListGuards
  - GetUserMenu                                     - TestGuard
  - CacheMenuForRole
  - InvalidateMenuCache

  MOBILE (6+)
  ----------
  - GetMobileNav
  - SetBottomTabs
  - ConfigureDrawer
  - SetQuickActions
  - HandleDeepLink
  - SyncNavState
```

@rottay/compliance (40+ Use Cases)

```
  KYC (10)                  AML (8)                 GDPR (10)
  -------                   -----                   ------
  - InitiateKYC             - ScreenTransaction     - RecordConsent
  - SubmitDocument          - CheckWatchlist        - WithdrawConsent
  - VerifyDocument          - FlagSuspicious        - ExportUserData
  - GetKYCStatus            - ReportSAR             - DeleteUserData
  - RequestAdditionalDocs   - GetRiskScore          - AnonymizeData
  - ApproveKYC              - SetTransactionLimit   - UpdateDataPolicy
  - RejectKYC               - ReviewAlert           - GetConsentHistory
  - ScheduleReview          - EscalateCase          - NotifyDataBreach
  - UpdateKYCLevel                                  - HandleDSAR
  - AuditKYCProcess                                 - SetRetentionPolicy

  AUDIT (7)                 CONSENT (5+)
  --------                  ----------
  - CreateAuditLog          - CreateConsentForm
  - QueryAuditLogs          - UpdateConsentForm
  - ExportAuditLogs         - GetActiveConsents
  - SetRetentionPolicy      - RevokeConsent
  - ArchiveOldLogs          - AuditConsentChanges
  - GenerateComplianceReport
  - ScheduleAudit
```

---

6. Domain Modules

> ***8 modules implementing vertical-specific business logic***

Domain Module Metrics

Module	Mutations	Queries	Total	Entities	Tests	LOC
@rottay/talent	55	34	**89**	11	265	107,788
@rottay/assessment	51	25	**76**	24	0	76,634
@rottay/ai-assistant	102	81	**183**	15	193	116,074
@rottay/web3	39	21	**60**	14	163	106,866
@rottay/bar	64	34	**98**	18	23	59,315
@rottay/staff	52	30	**82**	33	18	73,736
@rottay/ticketing	63	51	**114**	36	12	64,069
@rottay/payments	19	10	**29**	5	60	22,380
TOTAL	**445**	**286**	**731**	**156**	**734**	**626,862**

@rottay/talent

> ***Applicant Tracking System (ATS) with full recruitment lifecycle***

```
+-----------------------------------------------------------------+
|                          TALENT MODULE                           |
|        55 Mutations | 34 Queries | 89 Total | 11 Entities        |
|                   265 Tests | 107,788 LOC                        |
+-----------------------------------------------------------------+

  CANDIDATES (18)           JOBS (15)               APPLICATIONS (20)
  --------------            ------                  ---------------
  Mutations:                Mutations:              Mutations:
  - CreateCandidate         - CreateJob             - SubmitApplication
  - UpdateCandidate         - UpdateJob             - UpdateApplication
  - DeleteCandidate         - PublishJob            - WithdrawApplication
  - ImportCandidates        - UnpublishJob          - AdvanceStage
  - MergeCandidates         - CloseJob              - RejectApplication
  - AddCandidateNote        - ArchiveJob            - ScheduleInterview
  - TagCandidate            - DuplicateJob          - RequestDocuments
  - UpdateCandidateStatus   - SetJobPriority        - ScoreApplication
  - AssignRecruiter                                 - BulkUpdateStatus
                            Queries:                - AssignReviewer
  Queries:                  - GetJob                - AddApplicationNote
  - GetCandidate            - ListJobs              - RequestReference
  - ListCandidates          - SearchJobs
  - SearchCandidates        - GetJobStats           Queries:
  - GetCandidateHistory     - GetJobApplicants      - GetApplication
  - GetCandidatePipeline    - GetJobPipeline        - ListApplications
  - ExportCandidates                                - GetApplicationHistory
                                                    - GetPipelineMetrics

  INTERVIEWS (18)           OFFERS (18)
  -------------             ---------
  Mutations:                Mutations:
  - ScheduleInterview       - CreateOffer
  - RescheduleInterview     - UpdateOffer
  - CancelInterview         - SendOffer
  - CompleteInterview       - WithdrawOffer
  - SubmitFeedback          - ExtendOffer
  - RequestAvailability     - NegotiateOffer
  - SendReminder            - AcceptOffer
  - AssignInterviewer       - RejectOffer
  - CreateInterviewKit      - CounterOffer

  Queries:                  Queries:
  - GetInterview            - GetOffer
  - ListInterviews          - ListOffers
  - GetInterviewFeedback    - GetOfferHistory
  - GetInterviewerLoad      - GetOfferMetrics
  - GetAvailableSlots       - CompareOffers
```

@rottay/assessment

> ***Candidate evaluation with rubrics, scoring, and calibration***

```
+-----------------------------------------------------------------+
|                        ASSESSMENT MODULE                         |
|        51 Mutations | 25 Queries | 76 Total | 24 Entities        |
|                      0 Tests | 76,634 LOC                        |
+-----------------------------------------------------------------+

  RUBRICS (16)              SCORABLES (20)          CALIBRATION (15)
  -----------               -------------           ---------------
  Mutations:                Mutations:              Mutations:
  - CreateRubric            - CreateScorable        - CreateCalibration
  - UpdateRubric            - UpdateScorable        - ScheduleCalibration
  - DeleteRubric            - DeleteScorable        - StartCalibration
  - CloneRubric             - AssignScorable        - SubmitCalibrationScore
  - AddCriteria             - SubmitScore           - CompleteCalibration
  - UpdateCriteria          - UpdateScore           - ResolveDiscrepancy
  - RemoveCriteria          - FinalizeScore         - AdjustScores
  - SetWeights              - RequestReview         - SetBenchmark
  - PublishRubric           - ApproveScore
  - ArchiveRubric           - RejectScore           Queries:
                            - RecalibrateScore      - GetCalibration
  Queries:                                          - ListCalibrations
  - GetRubric               Queries:                - GetCalibrationResults
  - ListRubrics             - GetScorable           - GetDiscrepancies
  - GetRubricUsage          - ListScorables         - GetCalibrationStats
  - CompareRubrics          - GetScoreHistory
                            - GetAverageScores
                            - GetScorerStats

  ANALYTICS (25)
  -------------
  Mutations:                Queries:
  - GenerateReport          - GetAssessmentMetrics
  - ExportAnalytics         - GetScorerPerformance
  - ScheduleReport          - GetRubricEffectiveness
  - ConfigureDashboard      - GetPassRates
                            - GetTimeToComplete
                            - GetScoreDistribution
                            - GetInterRaterReliability
                            - TrendAnalysis
                            - BenchmarkComparison
```

@rottay/ai-assistant

> ***Multi-provider AI integration with conversation management and usage tracking***

```
+-----------------------------------------------------------------+
|                       AI-ASSISTANT MODULE                        |
|       102 Mutations | 81 Queries | 183 Total | 15 Entities       |
|                   193 Tests | 116,074 LOC                        |
+-----------------------------------------------------------------+

  SUPPORTED PROVIDERS
  +--------+---------+--------+---------+--------+------+--------+
  | OpenAI | Anthropic|  Groq  |ElevenLabs|Deepgram| Vapi | Retell |
  | GPT-4  |  Claude  |  LLaMA |   TTS   |  STT   | Voice | Voice |
  +--------+---------+--------+---------+--------+------+--------+

  PROVIDERS (35)            CONVERSATIONS (45)      MODELS (30)
  -------------             -----------------       ----------
  Mutations:                Mutations:              Mutations:
  - RegisterProvider        - CreateConversation    - RegisterModel
  - UpdateProvider          - UpdateConversation    - UpdateModel
  - DeleteProvider          - DeleteConversation    - DeprecateModel
  - SetProviderCredentials  - AddMessage            - SetDefaultModel
  - TestProviderConnection  - UpdateMessage         - ConfigureModel
  - SetProviderPriority     - DeleteMessage         - SetModelLimits
  - EnableProvider          - ClearConversation     - CloneModelConfig
  - DisableProvider         - ArchiveConversation   - SetFallbackModel
  - SetRateLimits           - ForkConversation      - BenchmarkModel
  - ConfigureFallback       - MergeConversations
  - SetCostLimits           - SummarizeConversation Queries:
  - RotateAPIKey            - RegenerateResponse    - GetModel
                            - StreamResponse        - ListModels
  Queries:                  - CancelGeneration      - GetModelCapabilities
  - GetProvider             - SetSystemPrompt       - GetModelPricing
  - ListProviders           - AddContext            - CompareModels
  - GetProviderStatus       - SetTemperature        - GetModelUsage
  - GetProviderUsage        - SetMaxTokens          - GetAvailableModels
  - GetProviderHealth       - EnableStreaming
  - GetProviderCosts        - AttachFile
  - CheckProviderLimits     - AttachImage

                            Queries:
                            - GetConversation
                            - ListConversations
                            - GetMessages
                            - SearchConversations
                            - GetConversationStats
                            - ExportConversation
                            - GetTokenCount

  USAGE (40)                EMBEDDINGS (15)         FINE-TUNING (18)
  ---------                 --------------          ---------------
  Mutations:                Mutations:              Mutations:
  - RecordUsage             - GenerateEmbedding     - CreateFineTune
  - SetUsageLimits          - BatchEmbed            - UploadTrainingData
  - ResetUsageCounter       - UpdateEmbedding       - StartFineTune
  - ConfigureBilling        - DeleteEmbedding       - CancelFineTune
  - SetAlertThresholds      - IndexEmbeddings       - DeployFineTune
  - GenerateInvoice                                 - RollbackFineTune
  - ApplyCredits            Queries:
                            - SearchByEmbedding     Queries:
  Queries:                  - GetSimilar            - GetFineTuneStatus
  - GetUsageStats           - GetEmbedding          - ListFineTunes
  - GetUsageByProvider      - ListEmbeddings        - GetTrainingMetrics
  - GetUsageByModel                                 - GetValidationResults
  - GetDailyUsage
  - GetMonthlyUsage
  - GetCostBreakdown
  - ForecastUsage
  - GetUsageTrends
```

@rottay/web3

> ***Blockchain integration across Polygon, Base, and BNB Chain***

```
+-----------------------------------------------------------------+
|                           WEB3 MODULE                            |
|        39 Mutations | 21 Queries | 60 Total | 14 Entities        |
|                   163 Tests | 106,866 LOC                        |
+-----------------------------------------------------------------+

  SUPPORTED CHAINS
  +----------+--------+---------+
  | Polygon  |  Base  |   BNB   |
  |  MATIC   |   ETH  |   BNB   |
  +----------+--------+---------+

  WALLETS (15)              TOKENS (12)             NFTs (12)
  -----------               ----------              --------
  Mutations:                Mutations:              Mutations:
  - CreateWallet            - DeployToken           - MintNFT
  - ImportWallet            - TransferToken         - TransferNFT
  - ConnectWallet           - ApproveSpender        - BurnNFT
  - DisconnectWallet        - RevokeApproval        - UpdateMetadata
  - SignMessage             - BurnToken             - ListNFT
  - SignTransaction         - MintToken             - DelistNFT
  - SetDefaultWallet                                - CreateCollection
                            Queries:
  Queries:                  - GetTokenBalance       Queries:
  - GetWallet               - GetTokenMetadata      - GetNFT
  - ListWallets             - ListTokens            - ListNFTs
  - GetBalance              - GetTransferHistory    - GetNFTMetadata
  - GetTransactionHistory   - GetTokenHolders       - GetCollectionStats
  - VerifyOwnership

  STAKING (10)              PAYMENTS (11)
  -----------               ------------
  Mutations:                Mutations:
  - StakeTokens             - CreatePaymentRequest
  - UnstakeTokens           - ProcessCryptoPayment
  - ClaimRewards            - RefundCryptoPayment
  - CompoundRewards         - SetPaymentAddress
  - SetAutoCompound         - ConfigureGasSettings

  Queries:                  Queries:
  - GetStakingPosition      - GetPaymentStatus
  - GetStakingRewards       - GetPaymentHistory
  - GetAPY                  - GetTransactionReceipt
  - GetStakingStats         - EstimateGas
                            - GetExchangeRate
                            - ValidateAddress
```

@rottay/bar

> ***Complete POS and inventory management for hospitality***

```
+-----------------------------------------------------------------+
|                           BAR MODULE                             |
|        64 Mutations | 34 Queries | 98 Total | 18 Entities        |
|                    23 Tests | 59,315 LOC                         |
+-----------------------------------------------------------------+

  PRODUCTS (20)             ORDERS (25)             INVENTORY (20)
  ------------              ----------              -------------
  Mutations:                Mutations:              Mutations:
  - CreateProduct           - CreateOrder           - AddInventoryItem
  - UpdateProduct           - UpdateOrder           - UpdateInventory
  - DeleteProduct           - CancelOrder           - AdjustStock
  - SetProductPrice         - AddOrderItem          - TransferStock
  - SetProductAvailability  - RemoveOrderItem       - ReceiveShipment
  - CreateCategory          - UpdateQuantity        - WriteOffStock
  - AssignCategory          - ApplyDiscount         - SetReorderPoint
  - SetModifiers            - RemoveDiscount        - CreateStockCount
  - CreateVariant           - SplitOrder            - SubmitStockCount
  - BulkUpdatePrices        - MergeOrders           - ReconcileStock
                            - HoldOrder
  Queries:                  - CloseOrder            Queries:
  - GetProduct              - VoidOrder             - GetInventoryLevel
  - ListProducts            - RefundOrder           - GetStockHistory
  - SearchProducts          - PrintReceipt          - GetLowStockItems
  - GetProductsByCategory                           - GetInventoryValue
  - GetBestSellers          Queries:                - GetStockMovements
                            - GetOrder              - ForecastDemand
                            - ListOrders
                            - GetOrderHistory
                            - GetDailySales
                            - GetOrderStats

  POS (18)                  SUPPLIERS (15)
  -------                   -------------
  Mutations:                Mutations:
  - OpenRegister            - CreateSupplier
  - CloseRegister           - UpdateSupplier
  - ProcessPayment          - DeleteSupplier
  - ProcessRefund           - CreatePurchaseOrder
  - ApplyCash               - ReceivePurchaseOrder
  - ApplyCard               - CancelPurchaseOrder
  - ApplySplit              - UpdateSupplierPricing
  - PrintReceipt
  - OpenCashDrawer          Queries:
  - RecordTip               - GetSupplier
                            - ListSuppliers
  Queries:                  - GetPurchaseOrders
  - GetRegisterStatus       - GetSupplierPerformance
  - GetCashDrawerBalance    - CompareSupplierPrices
  - GetShiftReport
  - GetDailyReport
  - GetPaymentBreakdown
```

@rottay/staff

> ***Workforce management with scheduling, credentials, and payroll***

```
+-----------------------------------------------------------------+
|                          STAFF MODULE                            |
|        52 Mutations | 30 Queries | 82 Total | 33 Entities        |
|                    18 Tests | 73,736 LOC                         |
+-----------------------------------------------------------------+

  STAFF (16)                SHIFTS (18)             CREDENTIALS (15)
  ---------                 ----------              ---------------
  Mutations:                Mutations:              Mutations:
  - CreateStaffMember       - CreateShift           - AddCredential
  - UpdateStaffMember       - UpdateShift           - UpdateCredential
  - DeactivateStaff         - DeleteShift           - DeleteCredential
  - AssignRole              - AssignShift           - VerifyCredential
  - UpdateContactInfo       - UnassignShift         - RenewCredential
  - SetAvailability         - SwapShifts            - ExpireCredential
  - UploadDocuments         - SplitShift            - UploadCredentialDoc
  - AssignManager           - PublishSchedule       - RequestVerification
                            - RequestTimeOff
  Queries:                  - ApproveTimeOff        Queries:
  - GetStaffMember          - DenyTimeOff           - GetCredential
  - ListStaff                                       - ListCredentials
  - SearchStaff             Queries:                - GetExpiringCredentials
  - GetStaffSchedule        - GetShift              - GetCredentialHistory
  - GetStaffHistory         - ListShifts            - ValidateCredentials
                            - GetWeeklySchedule
                            - GetStaffAvailability
                            - GetOpenShifts
                            - GetShiftConflicts

  TIME (16)                 PAYROLL (17)
  --------                  -----------
  Mutations:                Mutations:
  - ClockIn                 - CreatePayPeriod
  - ClockOut                - CalculatePayroll
  - StartBreak              - ApprovePayroll
  - EndBreak                - ProcessPayroll
  - EditTimeEntry           - AdjustPayment
  - ApproveTimeEntry        - AddBonus
  - RejectTimeEntry         - AddDeduction
  - SubmitTimesheet         - GeneratePaystub
                            - ExportPayroll
  Queries:
  - GetTimeEntry            Queries:
  - ListTimeEntries         - GetPayroll
  - GetTimesheetSummary     - GetPaystub
  - GetOvertimeReport       - GetPayrollHistory
  - GetAttendanceReport     - GetEarningsReport
                            - GetDeductionsReport
                            - GetTaxReport
                            - GetLaborCostReport
```

@rottay/ticketing

> ***Complete event ticketing platform with venues, live events, and resale***

```
+-----------------------------------------------------------------+
|                        TICKETING MODULE                          |
|       63 Mutations | 51 Queries | 114 Total | 36 Entities        |
|                    12 Tests | 64,069 LOC                         |
+-----------------------------------------------------------------+

  EVENTS (22)               TICKETS (24)            VENUES (18)
  ----------                -----------             ----------
  Mutations:                Mutations:              Mutations:
  - CreateEvent             - CreateTicketType      - CreateVenue
  - UpdateEvent             - UpdateTicketType      - UpdateVenue
  - PublishEvent            - DeleteTicketType      - DeleteVenue
  - UnpublishEvent          - IssueTicket           - AddSection
  - CancelEvent             - TransferTicket        - UpdateSection
  - PostponeEvent           - CancelTicket          - DeleteSection
  - RescheduleEvent         - RefundTicket          - AddSeatMap
  - CloneEvent              - ValidateTicket        - UpdateSeatMap
  - SetEventCapacity        - CheckInTicket         - SetSectionCapacity
  - AddEventImage           - UndoCheckIn           - BlockSeats
  - SetTicketPricing        - ReissueTicket         - UnblockSeats
                            - BulkIssueTickets
  Queries:                  - SetTicketLimit        Queries:
  - GetEvent                                        - GetVenue
  - ListEvents              Queries:                - ListVenues
  - SearchEvents            - GetTicket             - GetSeatMap
  - GetEventStats           - ListTickets           - GetAvailableSeats
  - GetEventAttendees       - GetTicketHistory      - GetSectionStats
  - GetEventRevenue         - ValidateTicketCode
  - GetEventCapacity        - GetTicketHolder
  - GetUpcomingEvents       - GetSalesReport
                            - GetCheckInStats

  LIVE (20)                 RESALE (15)             ANALYTICS (15)
  --------                  ----------              -------------
  Mutations:                Mutations:              Mutations:
  - StartLiveEvent          - ListTicketForResale   - GenerateEventReport
  - EndLiveEvent            - UpdateResaleListing   - ScheduleReport
  - PauseLiveEvent          - RemoveResaleListing   - ExportAnalytics
  - ResumeLiveEvent         - PurchaseResaleTicket  - ConfigureDashboard
  - UpdateAttendeeCount     - SetResalePriceLimit
  - SendLiveNotification    - ApproveResaleListing  Queries:
  - TriggerEmergencyAlert   - RejectResaleListing   - GetSalesAnalytics
  - UpdateEventStatus                               - GetRevenueAnalytics
  - LogIncident             Queries:                - GetAttendanceAnalytics
                            - GetResaleListings     - GetCheckInAnalytics
  Queries:                  - GetResaleHistory      - GetResaleAnalytics
  - GetLiveStatus           - GetResaleStats        - GetTrendAnalysis
  - GetLiveAttendance       - GetMarketPrice        - GetDemographicData
  - GetLiveIncidents        - SearchResaleTickets   - GetRevenueForcast
  - GetLiveMetrics                                  - CompareEvents
  - GetQueueStatus                                  - GetSeasonalTrends
  - GetEntryPoints
```

@rottay/payments

> ***Multi-provider payment processing with fiat and crypto support***

```
+-----------------------------------------------------------------+
|                         PAYMENTS MODULE                          |
|        19 Mutations | 10 Queries | 29 Total | 5 Entities         |
|                    60 Tests | 72,380 LOC                         |
+-----------------------------------------------------------------+

  SUPPORTED PROVIDERS
  +--------+-----------+---------+---------+------+------+---------+
  | Stripe | MercadoPago| MoonPay | Transak | Ramp | Wyre | Sardine |
  +--------+-----------+---------+---------+------+------+---------+
  | Banxa  |  Simplex  | Unlimit | Onramp  | Kado | Topper | +4 more |
  +--------+-----------+---------+---------+------+------+---------+

  PAYMENTS (8)              SUBSCRIPTIONS (6)       REFUNDS (5)
  -----------               ----------------        ----------
  Mutations:                Mutations:              Mutations:
  - CreatePayment           - CreateSubscription    - RequestRefund
  - ProcessPayment          - UpdateSubscription    - ApproveRefund
  - CapturePayment          - CancelSubscription    - ProcessRefund
  - VoidPayment             - PauseSubscription     - DenyRefund
                            - ResumeSubscription
  Queries:                                          Queries:
  - GetPayment              Queries:                - GetRefund
  - GetPaymentStatus        - GetSubscription       - GetRefundStatus
  - ListPayments            - ListSubscriptions
  - GetPaymentMethods       - GetSubscriptionHistory

  PAYOUTS (5)               CRYPTO (5)
  ----------                ---------
  Mutations:                Mutations:
  - CreatePayout            - ProcessCryptoPayment
  - ProcessPayout           - ConvertCrypto
  - CancelPayout            - WithdrawCrypto

  Queries:                  Queries:
  - GetPayout               - GetCryptoRates
  - ListPayouts             - GetCryptoBalance
```

---

7. Design System

> ***Full frontend customization from the portal — no code required***

Rottay includes a complete design system that allows white-label customers to fully customize their frontend appearance without writing code.

Architecture

```
+=====================================================================+
|                      DESIGN SYSTEM ARCHITECTURE                       |
+=====================================================================+
|                                                                       |
|                          ┌──────────┐                                 |
|                          │  PORTAL  │                                 |
|                          │ Branding │                                 |
|                          │ Settings │                                 |
|                          └────┬─────┘                                 |
|                               │                                       |
|              ┌────────────────┼────────────────┐                      |
|              │                │                │                      |
|              ▼                ▼                ▼                      |
|      ┌────────────┐    ┌──────────┐    ┌──────────┐                  |
|      │   TITAN    │    │  HERMES  │    │  APOLLO  │                  |
|      │ Ant Design │    │ Tailwind │    │ Vanilla  │                  |
|      │ Enterprise │    │  Modern  │    │ Minimal  │                  |
|      └─────┬──────┘    └────┬─────┘    └────┬─────┘                  |
|            │                │               │                        |
|            └────────────────┼───────────────┘                        |
|                             │                                        |
|                        ┌────┴─────┐                                  |
|                        │  THEMES  │                                  |
|                        │ Corporate│                                  |
|                        │ BitHire  │                                  |
|                        │ Minimal  │                                  |
|                        │ Custom...│                                  |
|                        └──────────┘                                  |
|                                                                       |
+=====================================================================+
```

4 UI Engines

Engine	Based On	Best For	Features
Titan	Ant Design	Enterprise apps	Feature-rich, comprehensive components
Hermes	Tailwind/DaisyUI	Modern startups	Lightweight, utility-first, fast
Apollo	Vanilla HTML/CSS	Minimal apps	No framework dependency, accessible
Athena	Pluggable	Custom needs	Bring your own component library

Theme Presets

Theme	Style	Primary Color	Best For
Foundation	Base	-	Extended by all themes
Corporate	Professional	#1E3A5F	B2B, Enterprise
BitHire	LinkedIn-style	#0A66C2	Recruiting, HR
Minimal	Notion-style	#37352F	Productivity, Docs

Portal Customization API

From the admin portal, tenants can customize:

Branding

- Main logo, dark mode logo, favicon
- App icons (192x192, 512x512)
- Login background image

Colors

- Primary, secondary, accent colors
- Success, warning, error states
- Background colors (light/dark mode)
- Text colors (primary, secondary, muted)

Typography

- Font families (heading, body, monospace)
- Base font size, weights, line height
- Letter spacing

Email Templates

- Header logo, footer text
- Social links (Twitter, LinkedIn, etc.)
- Custom colors for email campaigns

ThemeProvider Usage

```
import { ThemeProvider } from "@rottay/design-system";

// Basic usage with tenant resolution
<ThemeProvider tenant="acme-corp">
  <App />
</ThemeProvider>

// With runtime brand override
<ThemeProvider
  tenant="acme-corp"
  branding={{
    primaryColor: "#FF5733",
    accentColor: "#33FF57",
  }}
>
  <App />
</ThemeProvider>
```

Tenant Resolution

The design system automatically resolves tenant context through:

1. **Headers**: `x-tenant-id` header (server-side)
2. **Subdomains**: `acme.app.rottay.com`
3. **Custom Domains**: `recruiting.acme-corp.com`
4. **Fallback**: Defaults to `rottay` tenant

---

8. Web3 & Blockchain

> ***60 use cases | 3 networks | Complete Web3-as-a-Service***

The @rottay/web3 module provides enterprise-grade blockchain integration with multi-tenant support.

Supported Networks

```
+=====================================================================+
|                          BLOCKCHAIN NETWORKS                          |
+=====================================================================+
|                                                                       |
|   MAINNETS                              TESTNETS                      |
|   ────────                              ────────                      |
|   Polygon      (Chain ID: 137)          Polygon Amoy    (80002)       |
|   Base         (Chain ID: 8453)         Base Sepolia    (84532)       |
|   BNB Chain    (Chain ID: 56)           BNB Testnet     (97)          |
|                                                                       |
|   COMING SOON                                                         |
|   ───────────                                                         |
|   Solana (Non-EVM)                                                    |
|   Arbitrum, Optimism                                                  |
|                                                                       |
+=====================================================================+
```

Wallet Management

Wallet Type	Description	Use Case
Custodial	Platform-managed private keys	Simple user onboarding
Smart Wallet	Contract-based with session keys	Advanced UX, gasless txs
External	User-controlled, linked to platform	Crypto-native users
MPC	Multi-party computation	Enterprise security

Wallet Providers:

- ThirdWeb (primary)
- Privy
- Fireblocks

Token Management (ERC-20)

Deploy and manage custom tokens with configurable features:

```
// Deploy a custom token
const result = await deployToken.execute(ctx, {
  name: "Acme Rewards",
  symbol: "ACME",
  initialSupply: 1_000_000,
  features: {
    mintable: true,
    burnable: true,
    pausable: true,
    permit: true,      // Gasless approvals
    voting: false,
    flashMint: false,
  },
  maxSupply: 10_000_000,
});
```

NFT System

Badges (ERC-1155 - Semi-Fungible)

Type	Description
Achievement	Unlocked by completing actions
Milestone	Reached specific goals
Loyalty	Rewarded for engagement
Event	Attendance proof
Skill	Competency verification

Rarity Levels: Common, Rare, Epic, Legendary

Certificates (ERC-721 - Non-Fungible)

Type	Description
Completion	Course/training completion
Credential	Professional certification
License	Official authorization
Membership	Organization membership

Features:

- Soulbound option (non-transferable)
- Expiration dates
- On-chain verification codes
- Revocation support

Staking System

```
+===========================================================================+
|                           STAKING TIERS                                   |
+===========================================================================+
|                                                                           |
|   Tier        Lock Period    APY Range     Min Stake                      |
|   ────        ───────────    ─────────     ─────────                      |
|   Flexible    None           1-5%          None                           |
|   Bronze      30 days        5-8%          100 tokens                     |
|   Silver      90 days        8-12%         500 tokens                     |
|   Gold        180 days       12-18%        1,000 tokens                   |
|   Platinum    365 days       18-25%        5,000 tokens                   |
|   Diamond     730 days       25-35%        10,000 tokens                  |
|                                                                           |
|   Features: Compound rewards, early withdrawal penalties, per-tenant      |
|                                                                           |
+===========================================================================+
```

Fiat On/Off Ramp

Provider	Features
MoonPay	Card, bank transfer, Apple Pay
Transak	Global coverage, 100+ countries
Ramp	SEPA, PIX (Brazil), ACH (US)

---

9. Infrastructure & DevOps

> ***Production-ready infrastructure included with every deployment***

Multi-Tenant Architecture

```
+===========================================================================+
|                       MULTI-TENANT DATA ISOLATION                         |
+===========================================================================+
|                                                                           |
|   ┌──────────────────────────────────────────────────────────────────┐    |
|   │                         REQUEST FLOW                              │    |
|   │                                                                   │    |
|   │   Request → TenantResolver → TenantContext → All Queries         │    |
|   │                                                                   │    |
|   │   Every database query automatically filtered by tenant_id        │    |
|   │   Zero possibility of cross-tenant data leakage                  │    |
|   │                                                                   │    |
|   └──────────────────────────────────────────────────────────────────┘    |
|                                                                           |
|   TENANT ISOLATION FEATURES:                                              |
|   ├── tenant_id column on ALL tables                                     |
|   ├── TenantContext injected in every request                            |
|   ├── Custom domain support per tenant                                   |
|   ├── Per-tenant feature flags                                           |
|   ├── Per-tenant branding and themes                                     |
|   └── Per-tenant billing and subscriptions                               |
|                                                                           |
+===========================================================================+
```

Auto-Scaling & Load Balancing

Strategy	Description
Round Robin	Equal distribution
Weighted Random	Configurable weights per provider
Priority Based	Failover chain
Least Connections	Route to least busy

Resilience Features:

- Circuit breaker pattern
- Health check service
- Fallback orchestrator
- Provider scoring engine

Database Management

Feature	Implementation
ORM	Drizzle (type-safe)
Migrations	Versioned, automated
Seeds	18+ seed files for development
Database	PostgreSQL 16 optimized

Containerization

```
# Production-ready Docker setup
Services:
  - PostgreSQL 16 (persistent volumes)
  - Redis 7 (RDB + AOF persistence)
  - Typesense 26 (full-text search)
  - OpenObserve (observability)

Features:
  - Multi-stage Dockerfile
  - Alpine-based images (minimal attack surface)
  - Non-root user execution
  - Health checks built-in
  - Layer caching optimized
```

CI/CD Pipelines (GitHub Actions)

Pipeline	Purpose
`ci.yml`	Continuous integration
`cd-staging.yml`	Deploy to staging
`cd-production.yml`	Deploy to production (with gates)
`vercel-production.yml`	Vercel deployments

Production Gate: Requires typing "DEPLOY" for confirmation

Monitoring & Observability

```
+===========================================================================+
|                       OBSERVABILITY STACK                                 |
+===========================================================================+
|                                                                           |
|   ┌────────────┐   ┌────────────┐   ┌────────────┐                       |
|   │    LOGS    │   │  METRICS   │   │   TRACES   │                       |
|   │ OpenObserve │   │ Prometheus │   │   Jaeger   │                       |
|   └────────────┘   └────────────┘   └────────────┘                       |
|         │                │                 │                              |
|         └────────────────┼─────────────────┘                              |
|                          │                                                |
|                   ┌──────────────┐                                        |
|                   │   GRAFANA    │                                        |
|                   │  Dashboards  │                                        |
|                   └──────────────┘                                        |
|                                                                           |
|   Collector: OpenTelemetry                                                |
|   Alerting: Prometheus Alertmanager                                       |
|                                                                           |
+===========================================================================+
```

---

10. Metrics Summary

```
+===========================================================================+
|                         ROTTAY PLATFORM METRICS                           |
+===========================================================================+
|                                                                           |
|   +---------------------------+   +---------------------------+           |
|   |       USE CASES           |   |         ENTITIES          |           |
|   |          731              |   |           156             |           |
|   |   (445 mut + 286 qry)     |   |    (across 8 domains)     |           |
|   +---------------------------+   +---------------------------+           |
|                                                                           |
|   +---------------------------+   +---------------------------+           |
|   |      LINES OF CODE        |   |       TEST FILES          |           |
|   |         626,862           |   |           734             |           |
|   |     (production code)     |   |    (unit + integration)   |           |
|   +---------------------------+   +---------------------------+           |
|                                                                           |
|   +---------------------------+   +---------------------------+           |
|   |    PLATFORM MODULES       |   |     DOMAIN MODULES        |           |
|   |            7              |   |            8              |           |
|   |  (auth, tenancy, perms)   |   | (talent, payments, web3)  |           |
|   +---------------------------+   +---------------------------+           |
|                                                                           |
|   +---------------------------+   +---------------------------+           |
|   |    @rottay/core EXPORTS   |   |    SUPPORTED PROVIDERS    |           |
|   |          450+             |   |           20+             |           |
|   |  (errors, utils, types)   |   |   (AI, payments, chains)  |           |
|   +---------------------------+   +---------------------------+           |
|                                                                           |
+===========================================================================+
|  Architecture: Hexagonal + DDD + CQRS | Language: TypeScript 5.4+         |
|  Framework: Next.js 15 | Database: PostgreSQL 15 | Cache: Redis 7         |
+===========================================================================+
```

---

11. Code Patterns & Conventions

Naming Conventions

Element	Convention	Example
Folders	kebab-case	`use-cases/`, `staff-member/`
Files	kebab-case	`create-user.use-case.ts`
Interfaces	I + PascalCase	`IUserRepository`, `IPaymentGateway`
Types	PascalCase	`User`, `PaymentStatus`
Classes	PascalCase	`CreateUserUseCase`, `UserMapper`
Factories	make*	`makeUserRepository()`, `makePaymentService()`
Validators	*Schema	`createUserSchema`, `paymentInputSchema`
Constants	SCREAMING_SNAKE	`MAX_RETRY_ATTEMPTS`, `DEFAULT_PAGE_SIZE`
Enums	PascalCase	`UserStatus`, `PaymentType`

Result Pattern

```
// All use cases return Result<T, E>
import { Result, success, error, isSuccess, isError } from "@rottay/core";

// Success case
return success(user);

// Error case
return error(new ValidationError("Invalid email format"));

// Consuming results
const result = await createUser.execute(input);

if (isSuccess(result)) {
  const user = result.data;
  // Handle success
}

if (isError(result)) {
  const err = result.error;
  // Handle error
}
```

TenantContext Enforcement

```
// EVERY repository method receives TenantContext
// EVERY query is automatically scoped to tenant

interface IUserRepository {
  findById(ctx: TenantContext, id: string): Promise<User | null>;
  findAll(ctx: TenantContext, filters: UserFilters): Promise<User[]>;
  create(ctx: TenantContext, data: CreateUserData): Promise<User>;
}

// Implementation automatically applies tenant filter
async findById(ctx: TenantContext, id: string): Promise<User | null> {
  return this.db.query.users.findFirst({
    where: and(
      eq(users.id, id),
      eq(users.tenantId, ctx.tenantId),  // <-- Always enforced
      eq(users.isActive, true)            // <-- Soft delete filter
    )
  });
}
```

Soft Delete Pattern

```
// NEVER use DELETE - always set isActive = false

// Wrong
await this.db.delete(users).where(eq(users.id, id));

// Correct
await this.db.update(users)
  .set({
    isActive: false,
    updatedAt: new Date(),
    updatedBy: ctx.userId
  })
  .where(eq(users.id, id));
```

Standard Audit Fields

```
// Every entity includes these fields
interface AuditFields {
  id: string;           // UUID v7 (time-sortable)
  tenantId: string;     // Tenant isolation
  companyId: string;    // Company within tenant
  isActive: boolean;    // Soft delete flag
  createdAt: Date;      // Creation timestamp
  updatedAt: Date;      // Last modification
  createdBy: string;    // User who created
  updatedBy: string;    // User who last modified
}
```

---

12. Security Architecture

Multi-Tenant Isolation

```
+-------------------------------------------------------------------+
|                    TENANT ISOLATION LAYERS                         |
+-------------------------------------------------------------------+
|                                                                    |
|  Layer 1: Application Context                                      |
|  +------------------------------------------------------------+  |
|  |  TenantContext injected into every use case                |  |
|  |  - tenantId, companyId, userId extracted from JWT          |  |
|  |  - Context validated before any operation                   |  |
|  +------------------------------------------------------------+  |
|                              |                                     |
|  Layer 2: Repository Enforcement                                   |
|  +------------------------------------------------------------+  |
|  |  Every query automatically filtered by tenantId            |  |
|  |  - No way to bypass at repository level                    |  |
|  |  - Type system enforces TenantContext parameter            |  |
|  +------------------------------------------------------------+  |
|                              |                                     |
|  Layer 3: Database Row-Level Security                              |
|  +------------------------------------------------------------+  |
|  |  PostgreSQL RLS policies as final safeguard                |  |
|  |  - Even raw SQL queries are tenant-scoped                  |  |
|  |  - Audit logging for any cross-tenant access attempt       |  |
|  +------------------------------------------------------------+  |
|                                                                    |
+-------------------------------------------------------------------+
```

Encryption Strategy

Layer	Implementation	Scope
At Rest	AES-256-GCM	All database fields
In Transit	TLS 1.3	All API communication
Field-Level	@Encrypted decorator	PII, financial data, credentials
Secrets	AWS Secrets Manager	API keys, certificates
Tokens	JWT RS256	Session tokens, API tokens

```
// Field-level encryption with @Encrypted decorator
@Entity()
export class User {
  @Column()
  email: string;

  @Encrypted()  // <-- Automatically encrypted/decrypted
  @Column()
  ssn: string;

  @Encrypted()
  @Column()
  bankAccountNumber: string;
}
```

Compliance Certifications

Standard	Status	Scope
SOC 2 Type II	Compliant	Security, Availability, Confidentiality
HIPAA	Compliant	Healthcare data handling
GDPR	Compliant	EU data protection
ISO 27001	Compliant	Information security management
PCI-DSS	Level 1	Payment card data

---

13. Testing Strategy

Testing Pyramid

```
                     +-------+
                    /   E2E   \
                   /  Playwright \
                  / Happy Paths   \
                 +-------+-------+
                /                 \
               /    INTEGRATION     \
              /    Vitest + Real DB  \
             /   Repository + UseCase \
            +-----------+-----------+
           /                         \
          /           UNIT             \
         /           Vitest             \
        /  Domain Logic | Validators      \
       /     Transformers | Utilities       \
      +------------------+------------------+
```

Test Categories

Type	Tool	Scope	Database
Unit	Vitest	Pure functions, validators, transformers	None
Integration	Vitest	Repositories, use cases, services	Real PostgreSQL
E2E	Playwright	Full user workflows	Real PostgreSQL

Testing Philosophy: NO MOCKS

```
// WRONG - Mocking hides integration issues
const mockRepo = {
  findById: vi.fn().mockResolvedValue(mockUser),
  create: vi.fn().mockResolvedValue(mockUser),
};

// CORRECT - Real implementations, real database
describe("CreateUserUseCase", () => {
  let db: DrizzleClient;
  let userRepo: UserRepository;
  let useCase: CreateUserUseCase;

  beforeEach(async () => {
    db = await createTestDatabase();
    userRepo = new UserRepository(db);
    useCase = new CreateUserUseCase(userRepo);
  });

  afterEach(async () => {
    await cleanupTestDatabase(db);
  });

  it("should create a user with valid input", async () => {
    const result = await useCase.execute(ctx, validInput);

    expect(isSuccess(result)).toBe(true);

    // Verify in real database
    const dbUser = await userRepo.findById(ctx, result.data.id);
    expect(dbUser).not.toBeNull();
    expect(dbUser.email).toBe(validInput.email);
  });
});
```

Coverage Requirements

Metric	Minimum	Target
Line Coverage	80%	90%
Branch Coverage	75%	85%
Function Coverage	85%	95%
Critical Paths	100%	100%

Test File Distribution

Module	Test Files	Coverage
@rottay/talent	265	89%
@rottay/ai-assistant	193	84%
@rottay/web3	163	91%
@rottay/payments	60	88%
@rottay/bar	23	76%
@rottay/staff	18	72%
@rottay/ticketing	12	68%
@rottay/assessment	0*	-
TOTAL	**734**	**83%**

** Assessment module tests in development*

Appendix: Module Dependency Graph

```
+===========================================================================+
|                     MODULE DEPENDENCY GRAPH                                |
+===========================================================================+

                             @rottay/core
                                  |
         +------------+------------+------------+------------+
         |            |            |            |            |
         v            v            v            v            v
     @rottay/     @rottay/    @rottay/    @rottay/    @rottay/
       auth        tenancy     identity    perms       flags
         |            |            |            |            |
         +------------+-----+------+------------+------------+
                            |
                            v
                    DOMAIN MODULES
         +--------+--------+--------+--------+--------+
         |        |        |        |        |        |
         v        v        v        v        v        v
      talent  assessment   ai    web3     bar     staff
         |        |        |        |        |        |
         +--------+--------+---+----+--------+--------+
                               |
                               v
                        @rottay/payments
                               |
                               v
                        @rottay/ticketing
```

Built with precision. Engineered for scale. Ready for production.

626,862 lines of TypeScript | 731 use cases | 156 entities | 734 test files